Effects of coffee consumption in chronic hepatitis C: a randomized controlled trial.

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BACKGROUND: Coffee is associated with a reduced risk of hepatocellular carcinoma in patients with chronic C hepatitis. This prospective trial was aimed at assessing the mechanisms underlying coffee-related protective effects. METHODS: Forty patients with chronic hepatitis C were randomized into two groups: the first consumed 4 cups of coffee/day for 30 days, while the second remained coffee "abstinent". At day 30, the groups were switched over for a second month. RESULTS: At baseline, aspartate aminotransferase and alanine aminotransferase were lower in patients drinking 3-5 (Group B) than 0-2 cups/day (Group A) (56 ± 6 vs 74 ± 11/60 ± 3 vs 73 ± 7 U/L p=0.05/p=0.04, respectively). HCV-RNA levels were significantly higher in Group B [(6.2 ± 1.5) × 10(5)vs (3.9 ± 1.0) × 10(5)UI/mL, p=0.05]. During coffee intake, 8-hydroxydeoxyguanosine and collagen levels were significantly lower than during abstinence (15 ± 3 vs 44 ± 16 8-hydroxydeoxyguanosine/10(5)deoxyguanosine, p=0.05 and 56 ± 9 vs 86 ± 21 ng/mL, p=0.04). Telomere length was significantly higher in patients during coffee intake (0.68 ± 0.06 vs 0.48 ± 0.04 Arbitrary Units, p=0.006). Telomere length and 8-hydroxydeoxyguanosine were inversely correlated. CONCLUSION: In chronic hepatitis C coffee consumption induces a reduction in oxidative damage, correlated with increased telomere length and apoptosis, with lower collagen synthesis, factors that probably mediate the protection exerted by coffee with respect to disease progression.